

Mohammed Yaqoob

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EXPERIENCE

Software Engineering Intern

Jan 2026 – June 2026

Stalwart.ae · Dubai, UAE (Remote)

- Delivered production features across a React + Node.js stack, owning the full cycle from spec to live deployment in an agile sprint cadence.
- Integrated third-party payment and logistics APIs with retry queues and webhook signature validation, reducing failed transaction incidents by ~40%.
- Migrated untyped REST route handlers to strict Zod-validated schemas, eliminating a class of runtime input errors at the API boundary before they reached the database.
- Established the team's first test coverage on financial flows using Jest, covering critical payment paths end-to-end.

PROJECTS

Zebra Search | TypeScript, Node.js, SQLite Edge, React, Cloudflare Workers | GitHub

Feb – Apr 2026

- Built a complete search engine from first principles, strict-mode crawler, tokenized indexer, multi-factor ranking pipeline, and query UI with search frameworks.
- Crawler processed 50,000+ pages with enforced quality gates (minimum content length, keyword density thresholds, duplicate URL fingerprinting) to admit only high-signal documents into the index.
- Designed a hybrid retrieval model: BM25-style lexical scoring combined with embedding-based semantic similarity, weighted by freshness decay and domain authority signals.
- Query engine backed by Cloudflare D1 (edge SQLite); globally distributed deployment achieves sub-1s end-to-end query latency with no cold-start penalty.

Meridian | Python, FastAPI, PostgreSQL/pgvector, Redis, TypeScript/Next.js | GitHub

Jun – Aug 2026

- Built a RAG code-review system with tree-sitter chunking, storing 2048-dim embeddings in pgvector, and retrieving top-20 chunks via cosine similarity to ground LLM reviews in indexed codebase context.
- Automated PR reviews via HMAC-verified GitHub webhooks (<2s response), Redis deduplication (24h TTL), and RQ workers running a 4-stage pipeline: validation, retrieval, generation, and citation mapping.
- Streamed live pipeline progress over Redis Streams SSE with per-stage latency metrics and event replay so clients reconnecting mid-review resume without losing state.
- Enforced idempotent review generation using a (*repo_id, pr_number, head_sha*) dedup key and Redis locking, with batched ingest embedding (32 chunks/call) and checksum-based skip of unchanged files.

Sketch.io | React, TanStack Query, Node.js, WebSockets | GitHub

Nov – Dec 2025

- Built a real-time collaborative whiteboard where multiple users can draw on the same canvas simultaneously using WebSockets.
- Implemented canvas synchronization to keep drawings consistent across clients while handling concurrent updates from multiple users.
- Developed room-based session management, allowing users to create and join isolated whiteboard sessions without affecting other active rooms.

TECHNICAL SKILLS

Languages: TypeScript, Python, SQL, HTML/CSS

Frameworks & Runtimes: React, Next.js, Node.js, Express.js, FastAPI, Bun

Databases: PostgreSQL, MongoDB, Redis, MySQL, SQLite (Cloudflare D1)

Tools: Docker, Git, Turborepo, Cloudflare Workers, Prisma, Zod, WebSockets, SSE, Jest

AI / ML: LLM integration, embedding retrieval, RAG pipelines, vector scoring

EDUCATION

Shadan College of Engineering and Technology

Hyderabad, T.S.

Bachelor of Technology in Computer Science & Engineering

Nov 2022 – May 2026

ACHIEVEMENTS

Solved 350+ DSA problems on LeetCode and HackerRank.

Participated in 14+ contests, improving rating by 65%.